per cent of India's freshwater resources, any interference poses a strategic threat. "The proposed hydropower project has the potential to alter the flow dynamics of the Yarlung Tsangpo-Brahmaputra, affecting water availability, by influencing both groundwater and surface water levels," warns Genevieve Donnellon-May, a researcher at the Oxford Global Society.

For Assam, the implications for agriculture are particularly alarming. The river's nutrient-rich sediments sustain rice, tea and jute cultivation. Any disruption—excessive flooding or drought—could slash crop yields, threaten food security and financially cripple thousands of farmers. If upstream damming reduces sediment flow, it could accelerate riverbank erosion, degrade soil fertility and leave lasting ecological scars.

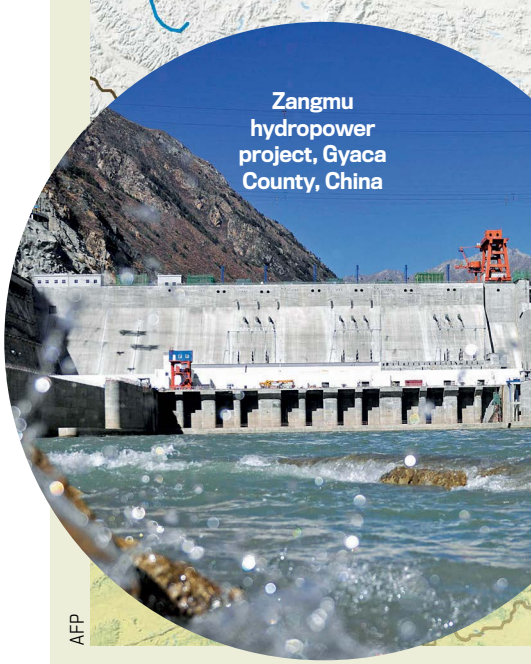
India's concerns over China's hydropower ambitions are well founded. In 2000, severe flooding in Arunachal Pradesh's Pasighat was linked to a dam collapse on Tibet's Yigong River. In 2012, the Siang River mysteriously ran dry, prompting then chief minister advisor Tako Dabi to blame China's dams. In 2016, China blocked the Xiabuqu River near India's border for the Lalho hydropower project, further raising suspicions. A year later, the Siang's waters turned black, igniting accusations against China. While Beijing dismissed the claims, satellite images later revealed that earthquakes on the Tibetan Plateau had triggered landslides, sending sediment downstream. "Storing water in a region with extensive mineral extraction—Tibet is home to over 100 exploited minerals—leads to contamination as mining runoff mixes with the water, forming black soot. Arunachal Pradesh has already seen instances of blackened water," says Srikanth Kondapalli, professor of Chinese Studies at Jawaharlal Nehru University. Adding to the uncertainty, open-source data from the Tibetan Autonomous Region suggests irregular river flow patterns in 2024, deviating from the trends of the past 25 years.

What raises even deeper concerns is China's broader ambition—to divert the Yarlung Tsangpo to its arid Xinjiang province. Gopal Dhawan, founder and chairman of the Dr Dhawan Academy of Geologists, and former CMD, Mineral Exploration and Consultancy Ltd and NHPC, warns that if China stores and redirects water through an inter-basin transfer scheme, "any projects we develop will face water shortages and be adversely affected." Though several experts rule out any such possibility, doubts persist. "The Medog dam appears to be a hydroelectric project rather than a water storage and diversion initiative. But given China's track record of undertaking large-scale infrastructure projects, such a possibility cannot be ruled out," says A.K. Bajaj, former chairman of the Central Water Commission. Amid these uncertainties, New Delhi has made its apprehensions clear to Beijing. "We have consistently raised concerns over China's mega river projects," says foreign ministry spokesperson Randhir Jaiswal, asserting India's lower riparian rights. He states that India will continue pressing China to safeguard downstream interests and "take necessary measures to protect our interests".



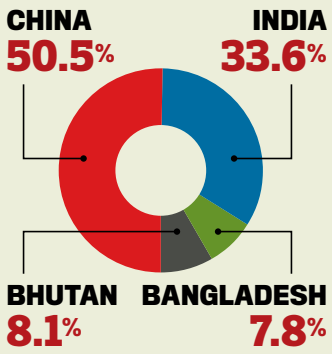
"This region is the most sediment-rich and sediment-producing area on Earth. As a result, dams here degrade faster, increasing the risk for downstream populations"

RUTH GAMBLE
Deputy Director (Research), La Trobe Asia, La Trobe University, Australia



A LONG COURSE

The Yarlung Tsangpo originates from the Angsi Glacier near Lake Manasarovar in western Tibet and flows into the Bay of Bengal. The Brahmaputra River extends across four countries



China's Water Hegemony

The Tibetan Plateau feeds 10 major river systems sustaining over a dozen Asian nations. Among them, two stand out for their geopolitical weight: the Mekong, which courses through Southeast Asia, and the Brahmaputra. China's aggressive dam-building along these rivers has triggered alarm over its push for hydro-hegemony, particularly given its refusal to sign water-sharing treaties. China's treatment of the